TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	77.0 ± 2.37
Correlation Stability	86.3 ± 3.32
Distribution Consistency	85.0 ± 2.49
Entropy Preservation	87.9 ± 1.30
Mean Z Score Drift	0.2 ± 0.03
Categorical Fidelity	82.8 ± 0.40
Privacy Risk	49.5 ± 1.63
Data Density	99.9 ± 0.17
Coverage Score	19.7 ± 4.05
Feature Accuracy	76.2 ± 1.89
Novelty Score	0.4 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	40.867	-3.879	8.7%	OK
TB	3.299	1.741	-1.558	47.2%	Drift
DB	1.486	0.740	-0.746	50.2%	Drift
Alkphos	290.576	220.461	-70.115	24.1%	Warn
Sgpt	80.714	41.839	-38.874	48.2%	Drift
Sgot	109.911	45.983	-63.928	58.2%	Drift
TP	6.483	6.696	+0.213	3.3%	OK
ALB	3.142	3.416	+0.274	8.7%	OK
A/G Ratio	0.947	1.015	+0.068	7.2%	OK
Selector	1.286	1.418	+0.132	10.3%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	77.0 ± 2.37
Correlation Stability (*2)	86.3 ± 3.32
Distribution Consistency (*3)	85.0 ± 2.49
Entropy Preservation (*4)	87.9 ± 1.30
Mean Z Score Drift (*5)	0.2 ± 0.03
Categorical Fidelity (*6)	82.8 ± 0.40
Privacy Risk (*7)	49.5 ± 1.63
Data Density (*8)	99.9 ± 0.17
Coverage Score (*9)	19.7 ± 4.05

Feature Accuracy (*10)	76.2 ± 1.89
Novelty Score (*11)	0.4 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.5521367521367522	0.6982905982905983	-0.005128205128205125	
F1	0.8108734553482932	0.6625211551885778	0.8041402198052368	-0.006733235543056615	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.5982905982905984	0.5717948717948718	0.012820512820512841	
F1	0.5587432119380917	0.6935466082363798	0.5805497657676927	0.021806553829601106	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6538461538461534	0.6871794871794872	0.000854700854700806	
F1	0.7971299319768493	0.7675439629346756	0.7994602968594264	0.002330364882577074	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6752136752136753	0.7068376068376068	0	
F1	0.8279091624529537	0.7888566141998537	0.8279091624529537	0	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6948717948717948	0.5743589743589743	0.6854700854700855	-0.009401709401709401	
F1	0.7973165851342273	0.66763342532134	0.7826736786097040	-0.014642906524523390	

Model: AB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.5649572649572653	0.6837606837606838	0.009401709401709401	
F1	0.7821343746683586	0.6742019839487653	0.7957320568898351	0.013597682221476459	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.5760683760683764	0.6948717948717948	-0.007692307692307675	
F1	0.7997903571549948	0.6739167882184752	0.7930596751351603	-0.006730682019834444	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6829059829059829	0.5470085470085471	0.6931623931623931	0.010256410256410286	
F1	0.7892724578593076	0.6518200236108534	0.7956609865027309	0.006388528643423585	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6615384615384616	0.529059829059829	0.6641025641025641	0.002564102564102555	
F1	0.7638877730901331	0.6318385511525673	0.7650158206543908	0.001128047564257639 4	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641026	0.5794871794871794	0.7025641025641025	1.1102230246251566e-17	
F1	0.809222862422675	0.6823447756990456	0.8088161834451283	-0.00040667897754671 5	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
2	39	Male	1.6	0.8	230	88	74	8.0	4.0	1.0
3	60	Male	1.8	0.5	201	45	25	3.9	1.7	0.7
4	6	Male	0.6	0.1	289	38	30	4.8	2.0	0.7
5	41	Male	1.2	0.5	246	34	42	6.9	3.4	0.97
6	64	Male	1.1	0.5	145	20	24	5.5	3.2	1.39
7	34	Female	1.2	0.4	210	45	50	6.5	3.0	1.1
8	52	Male	0.9	0.2	180	30	35	7.0	3.5	1.4
9	27	Female	0.5	0.1	150	25	28	5.8	2.8	0.9
10	41	Male	1.5	0.6	230	60	70	6.8	3.2	1.0
11	39	Female	0.8	0.3	190	50	55	7.2	3.6	1.2
12	45	Male	2.0	1.0	250	70	75	6.2	3.1	0.8
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	72	Male	0.7	0.2	185	16	22	7.3	3.7	1.0
2	42	Female	0.5	0.1	162	155	108	8.1	4.0	0.9
3	31	Male	1.3	0.5	184	29	32	6.8	3.4	1.0
4	70	Female	0.9	0.3	220	53	95	6.1	2.8	0.68
5	45	Male	2.2	0.8	209	25	20	8.0	4.0	1.0
6	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
7	29	Female	0.6	0.2	150	30	25	6.5	3.0	1.2
8	54	Male	1.2	0.4	220	50	60	7.0	3.5	1.0
9	38	Female	0.9	0.3	180	40	35	6.8	3.2	1.1
10	45	Male	2.1	0.5	200	45	55	7.5	3.8	1.3
11	62	Female	0.8	0.2	160	28	30	6.2	2.9	1.0
12	50	Male	1.5	0.7	210	55	50	7.1	3.6	1.1
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	26	Male	42.8	19.7	390	75	138	7.5	2.6	0.5
2	18	Male	1.3	0.7	316	10	21	6.0	2.1	0.5
3	46	Female	0.8	0.2	182	20	40	6.0	2.9	0.9
4	45	Male	0.7	0.2	180	18	58	6.7	3.7	1.2
5	36	Female	1.2	0.4	358	160	90	8.3	4.4	1.1
6	48	Female	1.4	0.8	621	110	176	7.2	3.9	1.1
7	34	Female	1.5	0.5	210	45	50	6.8	3.5	0.9
8	27	Male	0.9	0.3	180	30	40	6.0	2.8	0.7
9	52	Female	2.4	1.1	300	50	60	7.5	3.2	0.8
10	40	Male	3.2	1.5	400	60	70	8.0	4.0	1.0
11	29	Female	0.6	0.2	150	25	30	5.5	2.5	0.6
12	45	Male	1.8	0.9	250	40	50	6.5	3.0	0.7
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	33	Male	2.1	1.3	480	38	22	6.5	3.0	0.8
2	50	Male	7.3	3.7	92	44	236	6.8	1.6	0.3
3	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
4	60	Female	1.4	0.7	159	10	12	4.9	2.5	1.0
5	19	Male	1.4	0.8	178	13	26	8.0	4.6	1.3
6	70	Male	0.6	0.1	862	76	180	6.3	2.7	0.75
7	29	Female	1.2	0.4	150	30	25	6.0	3.5	1.2
8	55	Male	2.5	1.1	300	45	50	7.0	3.0	0.9
9	40	Female	0.9	0.3	200	20	30	5.5	2.8	0.7
10	34	Male	3.1	1.5	400	60	70	8.0	4.0	0.8
11	62	Female	1.5	0.6	180	35	40	6.8	3.2	1.1
12	48	Male	2.0	0.8	250	50	60	7.5	3.8	1.0
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	90	Male	1.1	0.3	215	46	134	6.9	3.0	0.7
2	60	Male	19.6	9.5	466	46	52	6.1	2.0	0.4
3	55	Male	0.9	0.2	116	36	16	6.2	3.2	1.0
4	48	Female	0.8	0.2	142	26	25	6.0	2.6	0.7
5	37	Male	0.7	0.2	176	28	34	5.6	2.6	0.8
6	60	Male	8.9	4.0	950	33	32	6.8	3.1	0.8
7	29	Female	1.2	0.4	185	28	30	6.5	3.2	0.8
8	52	Male	0.9	0.3	210	40	45	7.0	3.8	1.0
9	34	Female	3.5	1.2	250	50	60	6.8	3.5	0.9
10	41	Male	2.1	0.9	190	35	50	7.2	3.0	0.7
11	27	Female	0.6	0.2	160	20	25	6.3	3.1	0.8
12	38	Male	1.5	0.5	220	45	55	7.1	3.4	0.9
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	48	Female	0.8	0.2	142	26	25	6.0	2.6	0.7
2	33	Male	0.8	0.2	198	26	23	8.0	4.0	1.0
3	65	Male	0.9	0.2	170	33	66	7.0	3.0	0.75
4	47	Male	0.9	0.2	265	40	28	8.0	4.0	1.0
5	45	Male	3.2	1.4	512	50	58	6.0	2.7	0.8
6	85	Female	1.0	0.3	208	17	15	7.0	3.6	1.0
7	29	Female	0.9	0.3	150	22	18	6.5	3.0	0.8
8	34	Male	1.2	0.5	210	30	25	7.0	3.5	0.9
9	41	Female	0.8	0.2	175	28	20	6.8	3.2	1.0
10	55	Male	2.1	0.9	220	45	30	7.5	4.0	1.1
11	62	Female	1.5	0.6	180	35	22	6.7	3.1	0.85
12	27	Male	0.6	0.2	160	20	15	6.2	2.8	0.75
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Male	2.3	1.3	282	132	368	7.3	4.0	1.2
2	72	Male	1.7	0.8	200	28	37	6.2	3.0	0.93
3	66	Female	2.9	1.3	168	21	38	5.5	1.8	0.4
4	35	Female	0.9	0.2	190	40	35	7.3	4.7	1.8
5	60	Male	2.3	0.6	272	79	51	6.6	3.5	1.1
6	21	Male	1.0	0.3	142	27	21	6.4	3.5	1.2
7	29	Female	1.2	0.4	210	35	45	6.8	3.5	1.0
8	50	Male	0.8	0.3	180	40	50	7.2	3.8	1.2
9	38	Female	2.5	1.0	250	60	70	6.5	3.0	0.9
10	45	Male	1.1	0.5	200	55	65	7.0	3.4	1.1
11	62	Female	0.9	0.2	190	30	40	6.7	3.2	1.0
12	34	Male	1.5	0.6	220	45	55	7.5	3.6	1.3
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	48	Female	0.9	0.2	175	24	54	5.5	2.7	0.9
2	26	Male	1.9	0.8	180	22	19	8.2	4.1	1.0
3	60	Male	2.3	0.6	272	79	51	6.6	3.5	1.1
4	42	Female	0.8	0.2	195	18	15	6.7	3.0	0.8
5	35	Male	0.9	0.2	190	25	20	6.4	3.6	1.2
6	62	Male	0.7	0.2	162	12	17	8.2	3.2	0.6
7	29	Female	1.1	0.4	210	25	30	6.8	3.5	1.0
8	45	Male	2.5	1.0	300	50	60	7.2	3.8	2.0
9	52	Female	0.9	0.2	180	20	25	6.4	3.0	1.0
10	38	Male	1.2	0.5	220	30	35	7.0	3.4	1.0
11	60	Female	3.0	1.5	250	40	45	6.9	3.2	2.0
12	34	Male	0.8	0.3	190	22	28	6.5	3.1	1.0
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	58	Female	0.8	0.2	130	24	25	7.0	4.0	1.3
2	18	Male	0.8	0.2	282	72	140	5.5	2.5	0.8
3	38	Male	2.7	1.4	105	25	21	7.5	4.2	1.2
4	26	Female	0.6	0.2	142	12	32	5.7	2.4	0.75
5	28	Male	0.6	0.2	159	15	16	7.0	3.5	1.0
6	58	Female	0.7	0.1	172	27	22	6.7	3.2	0.9
7	29	Female	1.2	0.4	150	30	35	6.5	3.2	1.0
8	34	Male	0.9	0.3	200	45	50	7.0	4.0	1.2
9	42	Female	2.5	1.0	180	60	70	7.8	3.8	1.5
10	55	Male	0.6	0.1	220	25	30	5.5	2.9	0.9
11	63	Female	1.8	0.7	160	40	45	6.2	3.4	1.1
12	48	Male	0.7	0.2	190	35	38	6.8	3.6	1.0
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0
2	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
3	61	Male	0.8	0.2	163	18	19	6.3	2.8	0.8
4	32	Male	22.7	10.2	290	322	113	6.6	2.8	0.7
5	27	Male	1.2	0.4	179	63	39	6.1	3.3	1.1
6	29	Male	0.8	0.2	156	12	15	6.8	3.7	1.1
7	34	Male	1.0	0.4	210	45	50	6.5	3.2	0.8
8	29	Female	0.7	0.2	180	30	25	6.0	3.0	1.0
9	52	Male	2.5	1.0	300	50	60	7.0	3.5	0.9
10	40	Female	0.9	0.3	220	40	35	6.8	3.4	1.1
11	37	Male	1.3	0.5	250	55	45	6.4	3.1	0.7
12	45	Female	0.8	0.2	190	20	30	6.6	3.2	1.0
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3